



Microstructure and mechanical properties of friction stir processed Al–Mg–Si alloys dispersion-strengthened by nanosized TiB₂ particles



S.M. Ma^a, P. Zhang^a, G. Ji^b, Z. Chen^{a,*}, G.A. Sun^c, S.Y. Zhong^a, V. Ji^d, H.W. Wang^a

^a State Key Laboratory of Metal Matrix Composites, Shanghai Jiao Tong University, Shanghai 200240, PR China

^b Unité Matériaux et Transformations, CNRS UMR 8207, Université Lille 1, Villeneuve d'Ascq 59655, France

^c Institute of Nuclear Physics and Chemistry, Mianyang 621900, PR China

^d ICMCO, UMR CNRS 8182, Université Paris-Sud 11, Orsay Cedex 91405, France

ARTICLE INFO

Article history:

Received 11 June 2014

Received in revised form 9 July 2014

Accepted 10 July 2014

Available online 19 July 2014

Keywords:

Metal matrix composites

Grain boundaries

Mechanical properties

Microstructure

ABSTRACT

In this paper, we reported microstructure and mechanical properties of the in-situ TiB₂/Al–Mg–Si composites processed by friction stir processing (FSP). Compared with the initial state, the proper FSP conditions can enhance both the ultimate strength and the elongation of the composite samples obtained from the nugget zone. Detailed microstructure investigation has been performed by synchrotron X-ray diffraction, scanning and transmission electron microscopy and associated electron backscattered diffraction in order to reveal the mechanisms being responsible for the unusual mechanical behaviors. The results show that the initial composite has a grain size of 50–100 μm and the synthesized nanosized TiB₂ particles are almost agglomerated to form micrometer sized clusters at grain boundaries. Comparatively, after FSP, the nugget zone is characterized by fine and equiaxed recrystallized grains (1–5 μm in average grain size). The initial clusters are also broken up, while the nanosized TiB₂ particles are distributed much more uniformly in the matrix and act as effective pins to interact with dislocations. Hence, the significantly refined grains and the uniform distribution of the nanosized TiB₂ particles mainly contribute to the increase of both strength and ductility of the FSPed composites. The strengthening mechanisms are also discussed.

© 2014 Elsevier B.V. All rights reserved.

1. Introduction

It is very challenging to simultaneously increase strength and ductility of engineering materials. In the case of light-weight Al alloys, grain refinement by severe plastic deformation techniques, such as friction stir processing (FSP), equal channel angular pressing and high pressure torsion, can produce ultrafine-grained (UFG) alloys with increased strength. However, compared with their coarse-grained counterparts, ductility is often reduced due to low dislocation accumulation ability. One solution is then to introduce nanosized second-phase particles into the UFG matrix [1,2] since dislocation accumulation ability and resistance to dislocation slip can be effectively enhanced by the presence of the second-phase particles. As previously reported in the literatures, cryogenically rolled AA7075 [2], age-hardenable AA6063 [3], AA6082 alloys [4] and nano-particles reinforced Al composites [5] are all successful examples by using this solution.

In-situ particulate reinforced Al matrix composites, such as TiB₂/Al [6], TiC/Al [7] and ZrB₂/Al [8], have received considerable

attention over the past twenty years. They have demonstrated outstanding combination of mechanical properties, such as high specific strength, modulus, hardness and damping properties [8,9]. However, two common technical issues associated with the fabrication process can seriously limit the applications of the in-situ particulate reinforced metal matrix composites (MMCs). The one is particle clustering where small particles, especially those at the nanoscale in the melt, always tend to aggregate together in order to decrease free energy of the whole system. The harmful consequence is that it strongly degrades global mechanical properties of the composites, such as tensile strength, ductility, flow stress and fatigue resistance [10,11]; the other is cast defects remained in the composite ingot such as porosity.

Extremely high strain rate [12] and associated dynamical recrystallization generated during FSP can modify very much or even totally renew material microstructure, especially for the nanosized particles reinforced composites. FSP, as a new microstructure modification technique, has been found to be not only an efficient grain refinement technique for Al alloys [13] but also an effective dispersion method for particulate reinforced MMCs [14,15]. In addition, FSP is regarded as a promising approach to overcome the above mentioned problems of second-phase particle

* Corresponding author. Tel./fax: +86 21 5474 7597.

E-mail address: zhe.chen@sjtu.edu.cn (Z. Chen).

clustering and cast defects involved in the in-situ particulate reinforced MMCs [13]. Hence, FSP can be used to acquire an ideal microstructure containing nanosized second-phase particles uniformly dispersed in the UFG matrix for mechanical properties enhancement. However, very limited studies have been addressed so far in this field.

Among various ceramic reinforcements, TiB_2 particles possess strong covalent bonding, high strength and hardness, good damping [9], fatigue resistance [16] and neutron absorption properties, which make them a promising reinforcement candidate in the Al alloys for extreme-environment applications in the aerospace and nuclear industries. A recent study of Vijay and Murugan [17] has addressed a technical issue on the effect of tool pin profile on metallurgical and mechanical properties of the TiB_2/Al composites. This paper aims at the applicability of FSP technique for microstructure optimization of the in-situ $\text{TiB}_2/\text{Al-Mg-Si}$ composites in order to simultaneously improve strength and ductility. The effects of different FSP processing parameters (e.g. rotation speed, traverse speed and number of travel passes) on the microstructure and mechanical properties were studied. Statistical microstructural evaluation including grain and particle size distributions was examined in detail.

2. Experimental procedures

In this work, $\text{TiB}_2/\text{Al-Mg-Si}$ composites with 8 wt.% of TiB_2 particles were fabricated by in-situ mixed salt method as detailed in [6]. The reactive salts were K_2TiF_6 and KBF_4 (higher than 99.0% in purity). The composites were synthesized in a high-purity graphite crucible using an electrical resistance furnace at 900 °C. The pre-weighted mixture of K_2TiF_6 and KBF_4 was slowly added into the molten Al, being mechanically stirred at a rotation speed of 600 rpm for 15 min, and then the slag was poured out. Mg and Al–Si master alloys were subsequently added into the melt and homogenized for 10 min. The composite melt was cast in a graphite mould. Plates were cut from the equal-axial grain region of the solidified ingot and followed conventional homogenization, quenched in water and naturally aged for more than 96 h (referred to as the initial state hereafter). Chemical composition (in wt.%) of the Al–Mg–Si alloy, used as the matrix material, is given in Table 1.

The plates of the $\text{TiB}_2/\text{Al-Mg-Si}$ composite were then processed by FSP with the processing parameters given in Table 2. A taper round screwed stir rod used in this study is made of H13 steel, with the dimension of $\varnothing 4$ (top)/ $\varnothing 6$ (root) $\times$ 4 mm. The multi-passes FSP was performed in a reversal manner. The rotation and travel directions are shown in Fig. 1(a) and (b) presents the overview of a FSPed plate. A dog-bone like sample was directly cut from the nugget zone, which is parallel to the processed surface and along the travel direction. It has 1 mm in thickness and 10 mm in gauge length with 35 mm of overall length. Tensile tests were carried out on a Zwick/Roell machine at a strain rate of 10^{-3} s^{-1} . At least 3 samples were tested at each condition.

Microstructure of the composite samples before (i.e. initial) and after FSP was characterized by synchrotron X-ray diffraction (XRD), optical (OM), scanning electron microscopy (SEM) and transmission electron microscope (TEM). The synchrotron XRD measurements were performed on the BL14B beamline at the Shanghai synchrotron radiation facility (Shanghai, China). The specimens were irradiated by an incident beam of 10 keV photons (corresponding to a wavelength of $\lambda \sim 1.235 \text{ nm}$) with an approximate dimension of $0.3 \times 0.3 \text{ mm}^2$. In the FSP case, the beam was positioned at the center of the processed surface. High resolution diffraction patterns were recorded using a point detector.

An FEI NOVA NanoSEM 230 instrument equipped with an AZtec HKL Max electron backscattered diffraction (EBSD) detector was used for SEM examinations. SEM samples were prepared by mechanical and ion polishing. They were then observed under backscattered electron modes and analyzed by orientation imaging microscopy (OIM). EBSD Kikuchi patterns were post-treated using CHANNEL 5.0 software package in order to obtain information on grain and particle size distributions. OIM maps were recorded with step sizes of $0.02\text{--}0.5 \mu\text{m}$ in order to consider small recrystallized grains. Grain maps were reconstructed assuming that real grains contain two or more adjacent pixels with the same orientation and the grain boundary angles were higher than 15° (i.e. high angle grain boundaries (HAGB)). In addition, boundaries with misorientation angles between 2° and 15° were defined as

Table 2

Processing parameters of FSP used in this work.

Specimen	Rotation speed (rpm)	Traverse speed (mm/min)	Travel pass
1000-50-1p	1000	50	1
1000-200-1p	1000	200	1
1000-200-4p	1000	200	4
1500-50-1p	1500	50	1
1500-200-1p	1500	200	1

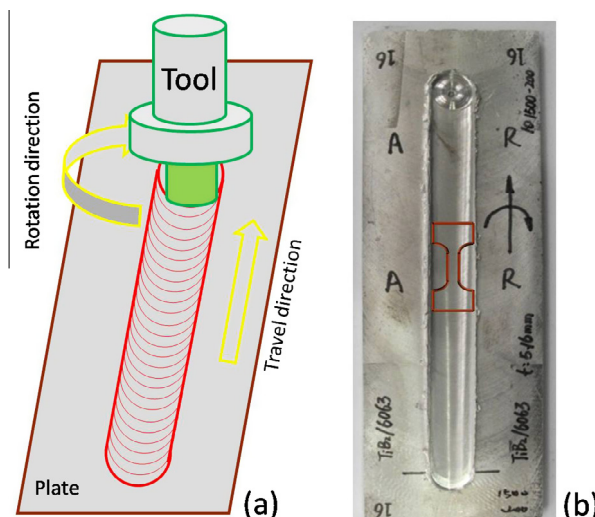


Fig. 1. (a) Schematic drawing showing rotation and travel directions during the FSP process; (b) plan-view photo of a FSPed composite sample. The location to acquire a dog-bone like sample for mechanical tests is noted. Letters A and R represent advancing and retreating sides, respectively.

low angle grain boundaries (LAGB) while spurious boundaries with misorientation angles lower than 2° were not studied. In all the OIM maps presented in this paper, grains are colored according to their orientations, with red, green and blue levels proportional to the three Euler directions.

TEM examinations were performed by using a Philips CM30 transmission electron microscope, operated at 300 kV and equipped with a Nanomega Spinning Star precession system and a Noran energy-dispersive X-ray (EDX) spectrometry system. TEM thin foils were prepared by mechanical polishing followed by final ion milling. Thin foils of the FSPed samples were obtained from the same nugget zone where the dog-bone like samples were prepared. Microdiffraction, i.e. with a nearly parallel incident beam focused on the specimen with a spot size in the range $10\text{--}50 \text{ nm}$, was performed to acquire a single-crystal zone-axis pattern (ZAP) of the nanosized TiB_2 particles.

3. Results and discussion

3.1. Microstructure of the initial in-situ $\text{TiB}_2/\text{Al-Mg-Si}$ composite

The synchrotron XRD patterns of the initial sample are shown in Fig. 2(a) and (b). The peaks of Al and TiB_2 are indexed according to JCPDS: 04-0787 and JCPDS: 35-0741, respectively. The strongest (220) peak of Mg_2Si (JCPDS: 35-0773) is also visible while no other brittle phases like Al_3Ti and impurity are detected.

The OIM map of the initial composite has a low indexation rate of about 50%. As shown in Fig. 3(a) and (b), the black areas (i.e. with blurred EBSD patterns) are mainly located along the grain bound-

Table 1

Chemical composition of the AA6063 Al-alloy matrix.

Mg (wt.%)	Si (wt.%)	Fe (wt.%)	Cu (wt.%)	Mn (wt.%)	Zn (wt.%)	Cr (wt.%)	Al
0.9	0.45	<0.35	<0.10	<0.10	<0.10	<0.10	Balanced

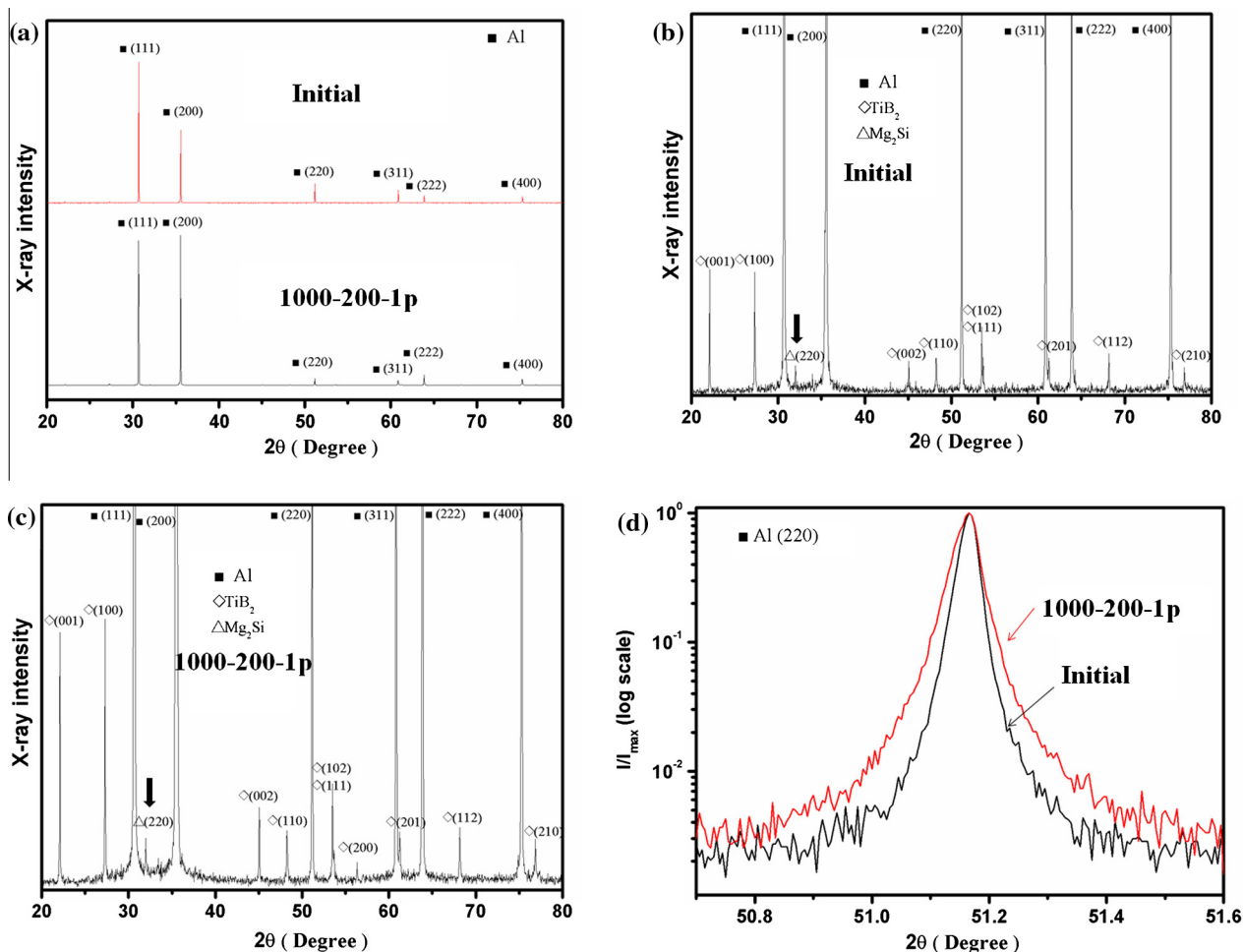


Fig. 2. (a) Synchrotron XRD patterns of the initial and FSPed (1000-200-1p, see Table 2) composite samples. Enlarged patterns of (b) the initial and (c) FSPed composite samples to highlight peaks with relatively low intensity, note that (b) and (c) are directly enlarged from (a) and Mg_2Si (220) peaks are arrowed; (d) high-resolution single peak profiles to compare Al (220) peak width of the initial and FSPed samples.

aries while only a few are inside the grains with a size in the range 50–100 μm . EDX analyses (Fig. 3(c) and (d)) indicate that these black areas are rich in Ti and B, which strongly suggests micrometer sized or micrometer sized clusters of TiB_2 particles. A relatively high indexation rate of 80–90% was achieved in the TiB_2 -free areas inside the grains, which suggests low density of substructures like dislocations and subgrains. The TEM image of the initial composite (in Fig. 4(a)) shows that the particles are almost cubic or polyhedral in morphology and in the size range from several tens to several hundreds of nanometers. As highlighted by Fig. 4(b), the nanosized particles are very much agglomerated and their nature identified by electron diffraction corresponds to the hexagonal TiB_2 phase. This result confirms that most of the TiB_2 particles are undetectable by EBSD analysis due to their tiny size, agglomeration and distribution overlapping with Al grain boundaries, which also cause the low indexation rate of EBSD. In addition, although depending on the crystallographic orientations, no visible dislocations inside the Al grains seem to indicate that dislocation density is low in the initial composite (Fig. 4(a)), being consistent with the above EBSD result.

3.2. Microstructure of the FSPed in-situ $\text{TiB}_2/\text{Al-Mg-Si}$ composites

3.2.1. Synchrotron XRD characterization

Qualitatively all the FSPed samples show almost the same synchrotron XRD patterns where the typical pattern of the composite processed in the 1000-200-1p condition is exhibited in Fig. 2(a).

The phase compositions of the FSPed sample, being the same as those of the initial sample, contain Al, TiB_2 and Mg_2Si phases (Fig. 2(c)). Furthermore, as exemplified in Fig. 2(d), the (220) Al peak width of the FSPed sample is clearly larger than that of the initial sample. Peak broadening after FSP indicates the introduction of higher dislocation density and/or smaller crystalline size.

3.2.2. OM, SEM and TEM characterization

Fig. 5(a) displays a cross-sectional OM image of the FSPed $\text{TiB}_2/\text{Al-Mg-Si}$ composite. The nugget zone appears as half-basin-shape. On the advancing side (AS), a sharp boundary is visible between the processed nugget zone and the unprocessed base material (BM). The typical onion ring pattern in the nugget zone is only clearly visible on the retreating side (RS). Some authors have previously reported a similar phenomenon in ZrB_2 and nanotube reinforced Al matrix composites and related it to the result of partial recrystallization in the nugget zone [8,18,19].

SEM image of the transition zone between the nugget zone and the BM is shown in Fig. 5(b). The areas with light-grey contrasts are related to the TiB_2 particles and their clusters. The boundary between the nugget zone and the BM is clearly identifiable. The band-like structures visible in the nugget zone are formed by redistributing the TiB_2 particles along the material flow direction during the FSP process. This fact indicates the thermodynamic cycles and reveals the shear and compress deformation generated by the rotating tool [18]. As shown in Fig. 5(c), in the BM, the TiB_2 particles are preferably agglomerated as clusters along the grain

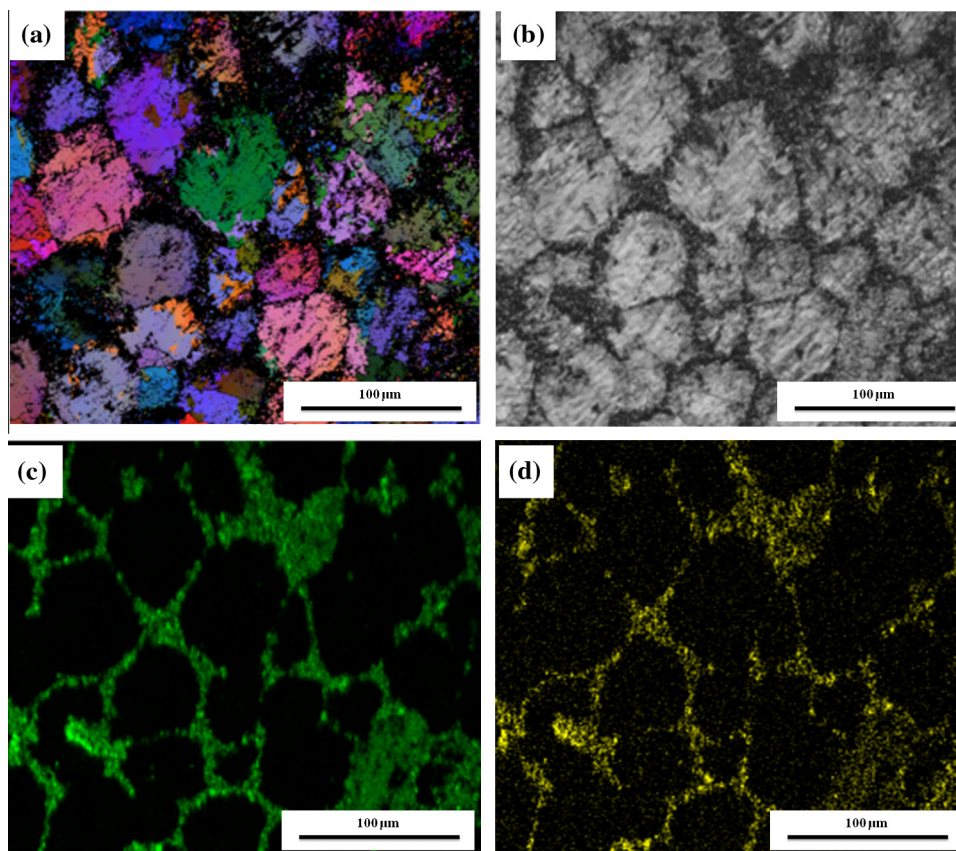


Fig. 3. (a) As-obtained EBSD/OIM map of the initial composite; (b) corresponding SEM image, (c) corresponding Ti and (d) B elemental EDX maps. Quantification of the OIM map was not performed due to the poor indexation rate (see text for more details).

boundaries and triple junctions. Comparatively, the TiB_2 particles in the nugget zone (Fig. 5(b)) are dispersed more homogeneously in the matrix and large clusters are hardly observed. This result is further confirmed by TEM examinations (in Fig. 4(d)). After FSP, the dispersed nanosized TiB_2 particles can clearly interact with dislocations inside the micrometer sized grain, which significantly contribute to ductility enhancement by the pinning effect.

3.2.3. Grain and particle size distributions characterized by SEM/EBSD

The indexation rate of around 80% is obtained in the FSPed sample processed in the 1000–200–1p condition. Consistent with the Ti and B distributions in Fig. 6(c) and (d), the undetected black areas in Fig. 6(a) are dispersed TiB_2 particles which are relatively uniformly dispersed in the matrix and show band fashion. Grain refinement is significant in the nugget zone after FSP processing. Fig. 6(e) shows that the average size of grains with HAGB is centred at around 2 μm , which is by one order smaller than the initial grain size between 50 and 100 μm (Fig. 3). The formation of fine equiaxed grains is due to heavy plastic deformation followed by dynamic recrystallization. During the FSP process, strain rate at the contact surface of tool pin and treated material can reach 80 s^{-1} [12], which is much larger than many traditional deformation approaches (0.1 – 10 s^{-1}). This can greatly contribute to grain refinement. Also as shown in Fig. 6(e), the distribution of grain size is much wider than the ideal log-normal grain size distribution, with the maximum grain size being around 8–10 times larger than the average. Wide grain size distribution is also strongly related to the distribution of TiB_2 particles. It is obvious that the strong stirring effect of a single-pass FSP can already be beneficial for the homogeneity of the particles distribution, although it is still not uniform. As shown in Fig. 6(a)–(d), the grain size is smaller along the TiB_2 distribution bands than that between the bands with less

TiB_2 particles. This is because the TiB_2 particles can restrict the dynamic recovery and slow down grain growth by pinning the movement of grain boundaries and sub-boundaries. Grain boundary misorientation distribution (in Fig. 6(f)) shows that the nugget zone contains more than 40 vol.% of LAGB, which is much larger than that of the FSPed Al alloys without particle reinforcement (10–20% [20]) due to the pinning effects of the nanosized TiB_2 particles.

The similar SEM/EBSD results of the FSPed sample processed in the 1000–200–4p condition are shown in Fig. 7. It can be seen that the distribution of TiB_2 particles is more homogenous in the 4-pass sample than that in the single-pass sample (Fig. 6) and no more band-like structures are observed here. Average grain size of this sample is around 3 μm which is slightly larger than that of the single-pass sample (Fig. 6(e)). However the number of grains with a size lower than 3 μm is significantly less than that of the single-pass sample. The more uniform distribution of TiB_2 particles in the matrix also leads to narrow grain size distribution. The maximum grain size is about 18 μm which is around 4–6 times larger than the average one. Fig. 7(f) shows the grain boundary misorientation distribution. Similar to Fig. 6(f) of the single-pass sample, the fraction of LAGB is more than 40%, but the fraction of small LAGB between 2° and 5° is slightly larger than that of the single-pass sample. This result further confirms the strong pinning effect of the dispersed nanosized TiB_2 particles on grain migration, especially on the LAGB.

3.3. Mechanical properties of the in-situ $\text{TiB}_2/\text{Al-Mg-Si}$ composites

Tensile test results containing yield strength (YS), ultimate tensile strength (UTS) and elongation of the initial and FSPed in-situ $\text{TiB}_2/\text{Al-Mg-Si}$ composites are given in Table 3. As can be

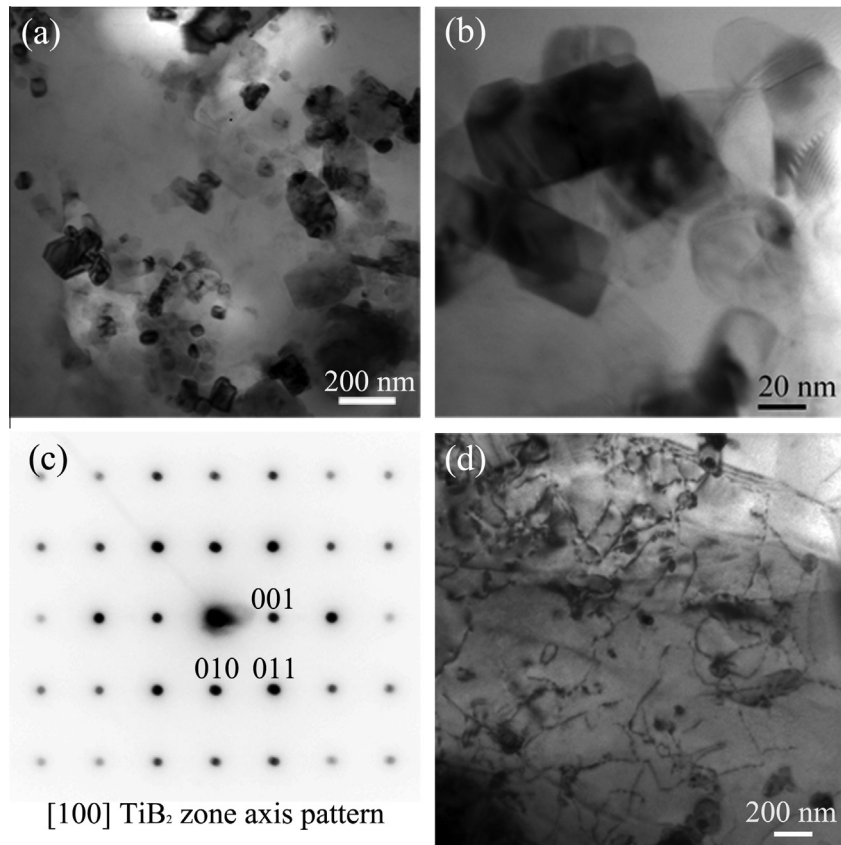


Fig. 4. (a) and (b) TEM bright-field images of the initial composite sample showing size, distribution and agglomeration of the TiB_2 particles; (c) $[100]$ ZAP from a nanosized particle corresponding to the TiB_2 phase and (d) TEM bright-field image of the FSPed composite sample (1000-200-1p, see Table 2) showing TiB_2 particles relatively uniformly dispersed inside a micrometer sized grain and their interaction with dislocations.

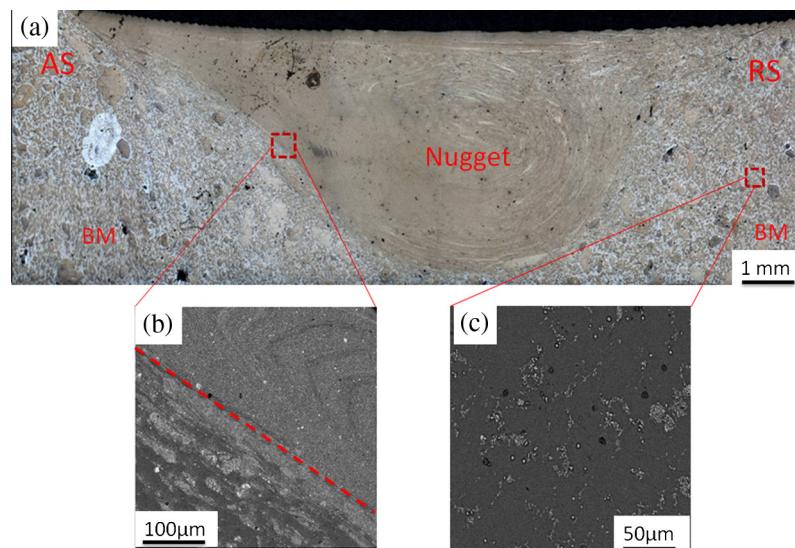


Fig. 5. (a) Typical OM image showing a cross section of the FSPed composite sample (1000-200-1p, see Table 2); SEM images of (b) the transition zone on the AS and (c) the BM away from the nugget zone. Dotted line in (b) marks the boundary between the nugget zone and the BM.

seen, only the 1500-50-1p condition is harmful for the YS, UTS and elongation of the FSPed composites while the other conditions are all beneficial for both strength and ductility. For example, UTS and elongation of the FSPed samples processed in the 1000-200-1p and 1500-200-1p conditions are 30–40% higher than those of the initial sample. The best FSP condition is 1000-200-4p where

the elongation is increased by more than 50% and the UTS by more than 40% with respect to those of the initial sample.

Both strength and ductility of the FSPed Al–Mg–Si Alloy are reduced primarily due to dissolution and/or coarsening of the hardening precipitates caused by high temperature locally raised during the FSP process [21]. Comparatively, in this case an

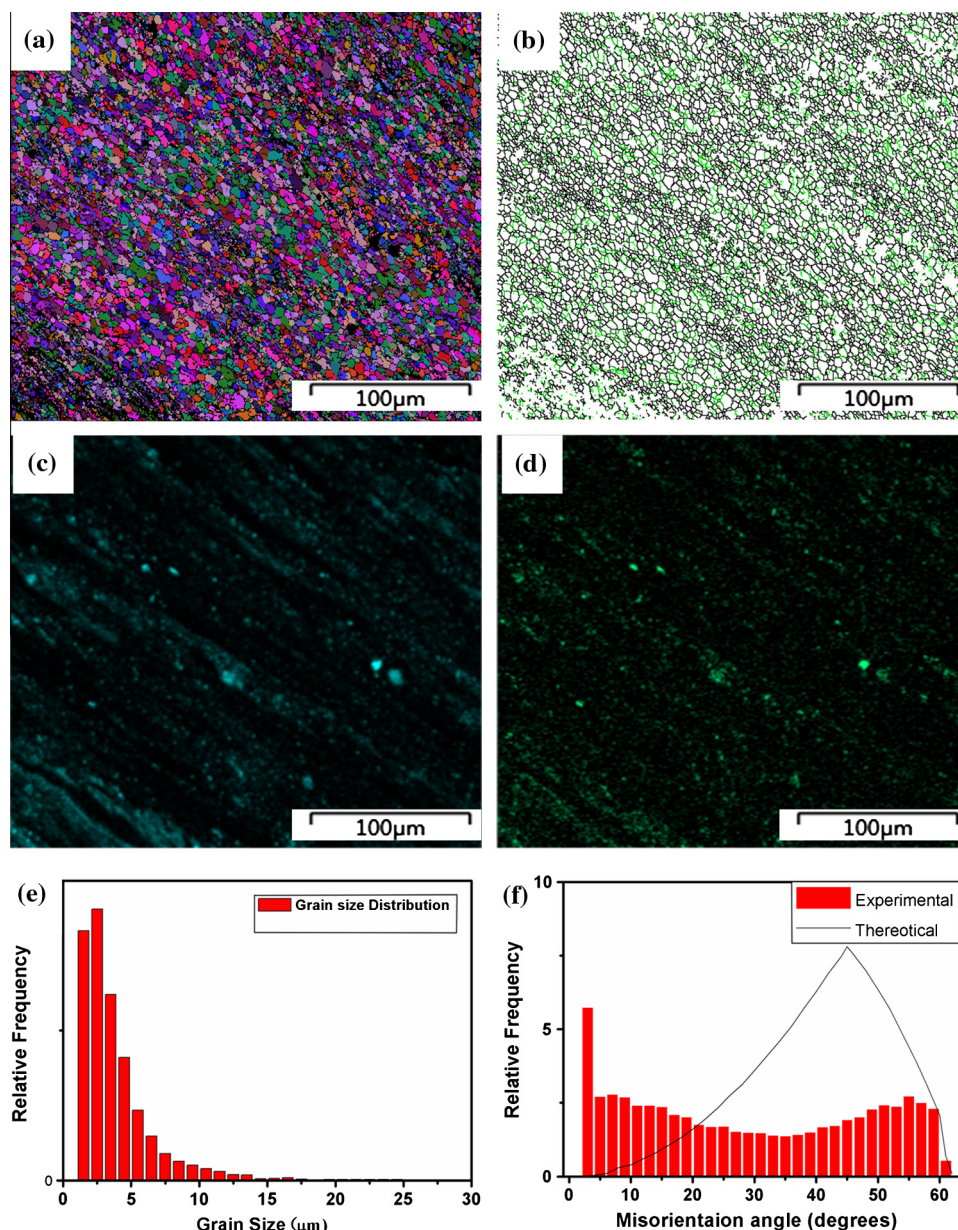


Fig. 6. (a) As-obtained OIM map of the FSPed composite sample (1000-200-1p, see Table 2); (b) corresponding treated OIM map showing grain boundary distribution; (c) corresponding Ti and (d) B elemental EDX maps; (e) statistical grain size and (f) grain boundary misorientation distributions obtained from the map shown in (a); high angle ($>15^\circ$) and low angle grain boundaries ($2 - 15^\circ$) in (b) are marked by black and green colors, respectively; additional black curve in (f) shows the theoretical distribution for a random grain assembly (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.).

Table 3

Mechanical properties of the in-situ $\text{TiB}_2/\text{AA6063}$ composites studied in this work.

Specimen	YS (MPa)	UTS (MPa)	Elongation (%)
Initial	128 ± 20	187 ± 20	5.7 ± 2
1000-50-1p	137 ± 15	245 ± 20	8.0 ± 2
1000-200-1p	140 ± 10	245 ± 20	11.0 ± 2
1000-200-4p	187 ± 15	275 ± 10	11.8 ± 3
1500-50-1p	111 ± 20	183 ± 20	3.6 ± 2
1500-200-1p	175 ± 20	261 ± 20	10.6 ± 2

introduction of the dispersed nanosized TiB_2 particles into the Al–Mg–Si matrix completely changes the situation. In terms of the microstructure examined in detail by synchrotron XRD, OM, SEM/EBSD and TEM, the enhancement mechanisms of strength as

well as ductility can be attributed to the refinement of grains in the matrix, the uniform distribution of the nanosized TiB_2 particles and the effect of different FSP parameters. However, the microstructure is significantly heterogeneous in the nugget zone throughout the sample thickness. Hence, the microstructure-properties relationship will only be discussed qualitatively as follows.

According to the well-known Hall–Petch relationship [22], YS is inversely proportional to the square root of average grain diameter. The statistical SEM/EBSD results shown in Figs. 3, 6 and 7 indicate that the average grain sizes of the FSPed composite samples (1–5 μm) are by one order smaller than that of the initial sample (50–100 μm). Hence, grain refinement is actually one of the main mechanisms for increasing the strength of the FSPed composite samples.

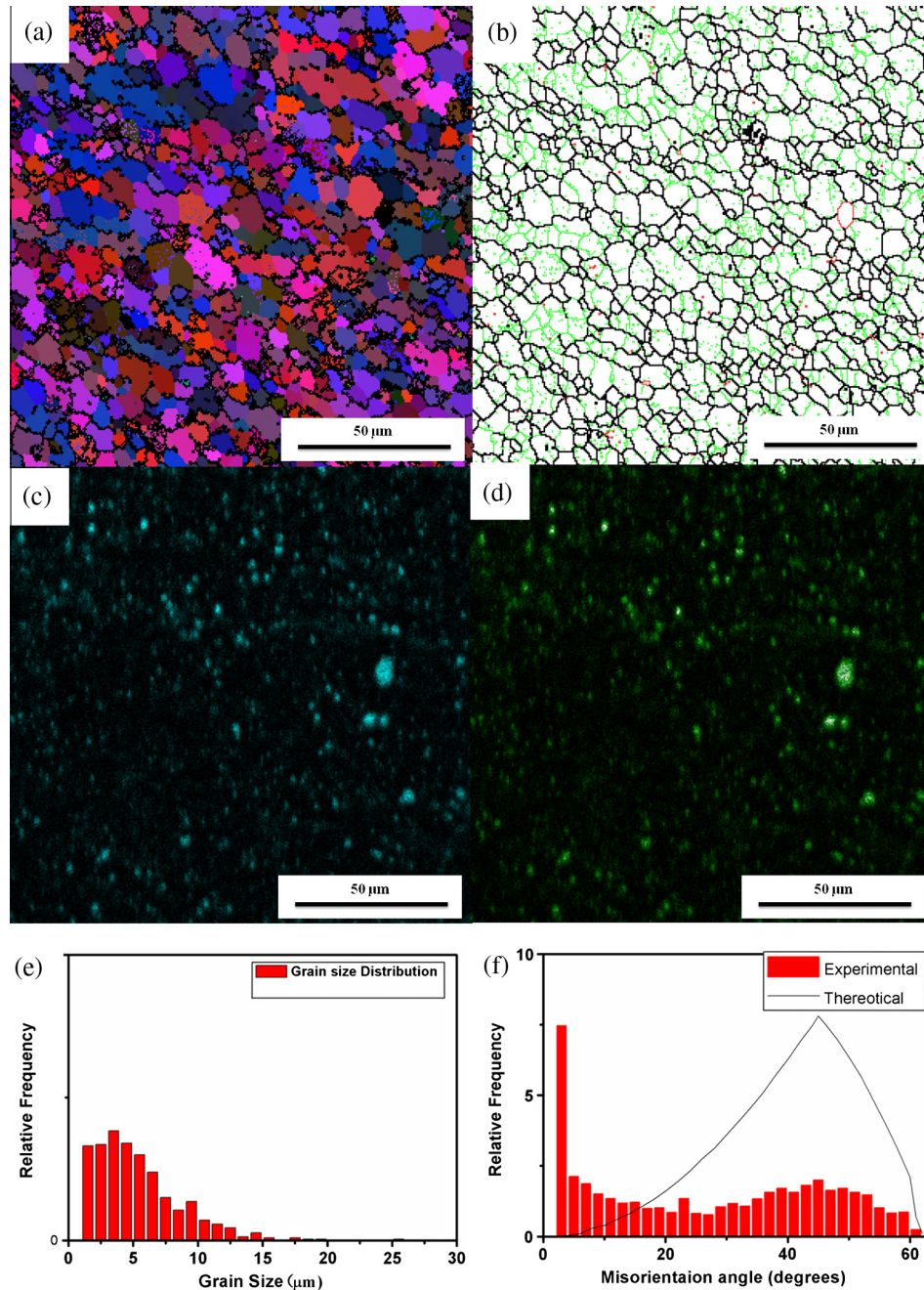


Fig. 7. (a) As-obtained OIM map of the other FSPed composite sample (1000-200-4p, see Table 2); (b) corresponding treated OIM map showing grain boundary distribution; (c) corresponding Ti and (d) B elemental EDX maps; (e) statistical grain size and (f) grain boundary misorientation distributions obtained from the map shown in (a); high angle (>15°) and low angle grain boundaries (2–15°) in (c) are marked by black and green colors, respectively; additional black curve in (f) shows the theoretical distribution for a random grain assembly (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.).

The uniform distribution of the nanosized TiB_2 particles plays another important role in both strength and ductility enhancement in the FSPed $\text{TiB}_2/\text{Al-Mg-Si}$ composites. According to the Orowan strengthening mechanism, strengthening of the composites is attributed to the interaction between the reinforcement particles and the motion of dislocations in the matrix. Normally the size of second-phase particles should be smaller than 1 μm to initiate this strengthening mechanism. In our case, the non-shearable TiB_2 particles of the size smaller than 500 nm act as pins to hinder the movement of dislocations. The general form to elucidate Orowan strengthening effect is given as the Orowan–Ashby equation (i.e. Eqs. (1a) and (1b)) [23]:

$$\Delta\sigma_{\text{Orowan}} = \frac{0.4MG_m b}{\pi\lambda} \times \frac{\ln\left(\sqrt{\frac{2}{3}}d_p/b\right)}{\sqrt{1-\nu}} \quad (1a)$$

$$\lambda = \sqrt{\frac{2}{3}}d_p \left(\sqrt{\frac{\pi}{4V_p}} - 1 \right) \quad (1b)$$

where M is the mean orientation factor ($M_{\text{Al}} = 3.06$); λ is the inter-particle spacing.

The micrometer sized TiB_2 particle clusters in the initial composite (Figs. 3–5) are broken up and uniformly dispersed in the composites after FSP (Figs. 4–7). As a result, the interval of the

TiB₂ particles decreases so that the Orowan strengthening effect is enhanced. In addition, the effective interaction between dislocations and dispersed nanosized TiB₂ particles increases the dislocation storage capability and thus improves the ductility of the composites. Our phenomenological model has recently suggested that a coefficient should be introduced to account for the effective particles content in the strengthening of the TiB₂/Al composites [24], since clustering of the reinforced particles could have little or even adverse effect on the strengthening of mechanical properties in the composites. Further homogenization of the nanosized TiB₂ particles distribution by 4-pass FSP can further reduce the interparticle spacing and also increase the effective TiB₂ particles content. It can be considered as the main reason why the strength of the 4-pass sample is better compared with those of the single-pass sample (Table 3).

In the Al–Mg–Si alloy, the presence of fine Mg₂Si precipitates is helpful for enhancing the strength of the alloy. Generally, it is well accepted in the MMCs that the reinforced particles should accelerate the precipitating kinetics in the matrix, because the addition of such particles can create more dislocations in the matrix [25,26]. This increase in dislocation density promotes dislocation-assisted diffusion of the alloying elements from the adjacent dislocation-lean areas of the matrix, resulting in solute enrichment in the interfacial region, thus making the compositional requirement for precipitation more favorable [27]. As a result, the nucleation and growth of nanosized precipitates may both be promoted, which can strengthen the composites in turn. As the TiB₂ particles are distributed more homogeneously in the FSPed samples, the particle stimulated precipitation distribution becomes more homogenous compared with that in the as-casted sample, which can further increase the relative contribution of Orowan strengthening effect.

Finally, FSP is a very complex process associated with a series of parameters such as temperature (*T*), strain (*ε*), and strain rate (*ε̇*). All of them would have important effects on microstructural evolution and mechanical properties. Large rotation speed increases the strain and strain rate while it may also raise the temperature, as shown in Eq. (2) [13]

$$\frac{T}{T_m} = K \left(\frac{\omega^2}{v \times 10^4} \right)^\alpha \quad (2)$$

where *T_m* is the melt point of Al alloys; *K* is a constant between 0.65 and 0.75; the exponent *α* is a constant between 0.04 and 0.06; *ω* is the rotation speed (rpm) and *v* is the traverse speed (mm/min).

As shown in Table 3, the 1500-50-1p sample shows even worse strength and ductility than the initial state. This indicates that too high rotation speed (*ω*)/traverse speed (*v*) ratio deteriorates the mechanical properties. The microstructure of this sample (not presented here) shows overheated structure and macroscopically visible pores which might be caused by too high temperature and over softening of the Al matrix. As the rotation speed (*ω*)/traverse speed (*v*) ratio decreases, such as in the 1500-200-1p and 1000-50-1p samples, both strength and ductility are very much improved. The comparison between 1000-200-1p and 1000-200-4p samples indicates the strain (*ε*) is another significant factor affecting the mechanical properties. However, since the strain of FSP is closely related to the temperature through changing the rotation speed (*ω*) and traverse speed (*v*), it is very difficult to give detailed analysis based on the existing results. More systematic studies will be carried out in the near future. The results obtained in this work show that multi-pass FSP processing with proper rotation speed/traverse speed ratio has high potential to enhance both strength and ductility of the composite.

4. Summary and conclusions

The initial plates of the in-situ TiB₂/Al–Mg–Si composites were processed by FSP in order to further improve their mechanical

properties. Microstructure of the initial and FSPed (with different FSP conditions) composites has been characterized by synchrotron XRD, OM, SEM/EBSD and TEM. The results show that the proper FSP conditions can uniformly distribute nanosized TiB₂ particles into the significantly refined grain matrix so that both strength and ductility of the FSPed composites in the nugget zone can be very much improved. The main conclusions are drawn below:

- (1) The grain size of the initial composite is in the range 50–100 μm while that of the FSPed composite is between 1 and 5 μm. The refined equiaxed grains are formed due to heavy plastic deformation followed by dynamic recrystallization. The micrometer sized clusters of the nanosized TiB₂ particles in the initial composite are broken up and distributed more uniformly in the FSPed composites. The presence of the well-dispersed nanosized TiB₂ particles is beneficial since they can stimulate the nucleation of recrystallized grains and in the same time restrict dynamic recovery and slow down the grain growth by pinning the movement of dislocations and grain boundaries, especially the LAGB.
- (2) Confirmed by tensile tests, both the strength and ductility of the composites are greatly enhanced by FSP. The improved strength is attributed to the refined grain size and uniformly dispersed nanosized TiB₂ particles in terms of the Hall–Petch relationship and Orowan strengthening mechanism, respectively. The dispersed nanosized TiB₂ particles also increase the dislocation storage capability to improve the ductility of the composites.
- (3) The FSP parameters significantly influence mechanical properties of the composites. The maximum temperature raised during the process goes down with the decrease of rotation speed/traverse speed ratio. As a result, both the strength and ductility are increased. In addition, multi-passes FSP can further enhance the strength of the composites.

Acknowledgements

This work is financially supported by the National Natural Science Foundation of China (Grant Nos. 51201099 and 51301108). The authors are grateful for the project 2013BB03 supported by NPL, CAEP. Many thanks are also due to the faculty members of BL14B beamline at the Shanghai Synchrotron Radiation Facility for their help on synchrotron experiments. The TEM facility in Lille (France) is supported by the Conseil Regional du Nord-Pas de Calais and the European Regional Development Fund (ERDF).

References

- [1] I. Sabirov, M.Y. Murashkin, R.Z. Valiev, *Mater. Sci. Eng. A* 560 (2013) 1–24.
- [2] Y.H. Zhao, X.Z. Liao, S. Cheng, E. Ma, Y.T. Zhu, *Adv. Mater.* 18 (2006) 2280–2283.
- [3] S.K. Panigrahi, R. Jayaganthan, *Metall. Mater. Trans. A* 41 (2010) 2675–2690.
- [4] S. Dadbakhsh, A. Karimi Taheri, C.W. Smith, *Mater. Sci. Eng. A* 527 (2010) 4758–4766.
- [5] C.M. Hu, C.M. Lai, X.H. Du, N.J. Ho, J.C. Huang, *Scr. Mater.* 59 (2008) 1163–1166.
- [6] D. Chen, Z. Chen, P. Zhang, Y. Zhang, N. Ma, H. Wang, *Adv. Eng. Mater.* 1–3 (2012) 1005–1010.
- [7] H.-J. Kim, J.-M. Lee, Y.-H. Cho, J.-J. Kim, S.-H. Kim, J.-C. Lee, *Korean J. Met. Mater.* 50 (2012) 677–684.
- [8] I. Dinaharan, N. Murugan, *Mater. Sci. Eng. A* 543 (2012) 257–266.
- [9] Y. Zhang, N. Ma, H. Wang, Y. Le, S. Li, *Scr. Mater.* 53 (2005) 1171–1174.
- [10] I.G. Watson, M.F. Forster, P.D. Lee, R.J. Dashwood, R.W. Hamilton, A. Chirazi, *Composites A* 36 (2005) 1177–1187.
- [11] A.R. Kennedy, S.M. Wvatt, *Compos. Sci. Technol.* 60 (2000) 307–314.
- [12] S. Mukherjee, A.K. Ghosh, *Mater. Sci. Eng. A* 527 (2010) 5130–5135.
- [13] R.S. Mishra, Z.Y. Ma, *Mater. Sci. Eng. R* 50 (2005) 1–78.
- [14] Y. Morisada, H. Fujii, T. Nagaoka, M. Fukusumi, *Mater. Sci. Eng. A* 433 (2006) 50–54.
- [15] X.G. Chen, M. da Silva, P. Gougeon, L. St-Georges, *Mater. Sci. Eng. A* 518 (2009) 174–184.

- [16] F. Wang, X. Meng, N. Ma, J. Xu, X. Li, H. Wang, J. Mater. Sci. 47 (2012) 3361–3366.
- [17] S.J. Vijay, N. Murugan, Mater. Des. 31 (2010) 3585–3589.
- [18] Q. Liu, L. Ke, F. Liu, C. Huang, L. Xing, Mater. Des. 45 (2013) 343–348.
- [19] D.-H. Choi, Y.-H. Kim, B.-W. Ahn, Y.-I. Kim, S.-B. Jung, Trans. Nonferr. Met. Soc. 23 (2013) 335–340.
- [20] K.A.A. Hassan, A.F. Norman, D.A. Price, P.B. Prangnell, Acta Mater. 51 (2003) 1923–1936.
- [21] Y. Sato, H. Kokawa, Metall. Mater. Trans. A 32 (2001) 3023–3031.
- [22] C.S. Pande, K.P. Cooper, Prog. Mater. Sci. 54 (2009) 689–706.
- [23] A. Sanaty-Zadeh, Mater. Sci. Eng. A 531 (2012) 112–118.
- [24] M. Wang, D. Chen, Z. Chen, Y. Wu, F. Wang, N. Ma, H. Wang, Mater. Sci. Eng. A 590 (2014) 246–254.
- [25] P.R.M.T. Das, S. Bandyopadhyay, J. Mater. Sci. 31 (1996) 5351–5361.
- [26] P.B.T.J.A. Doel, Compos. A Appl. Sci. Manuf. 27A (1996) 655–665.
- [27] P.B. Prangnell, W.M. Stobbs, Proceedings of the 7th International Conference on Composite Materials (ICCM7), Pergamon, 1990. pp. 573–678.